DualDistill: A Unified Cross-Modal Knowledge Distillation Framework for Camera-Based BEV Representation

Gaeun Kim*[1]
ge_kim@seoultech.ac.kr
Daeil Han*[1]
dihan@seoultech.ac.kr
Yeong Jun Koh[2]
yjkoh@cnu.ac.kr
Hanul Kim[1]
hukim@seoultech.ac.kr

[1] Seoul National University of Science and Technology, Seoul, South Korea
[2] Chungnam National University, Daejeon, South Korea

Abstract

Cross-modal knowledge distillation has drawn much attention to camera-based bird's-eye-view (BEV) models, aiming to narrow the performance gap with their LiDAR-based counterparts. However, distilliing knowledge from a LiDAR-based teacher is not easy due to the discrepancy between sensor modalities. In this work, we introduce DualDistill, a unified cross-modal knowledge distillation framework to address this challenge. We propose an attention-guided orthogonal alignment (AOA) to align student features with the teacher's representations while preserving useful information. This alignment is integrated into a multi-scale feature distillation with adaptive region weighting scheme. In addition, we introduce a cross-head response distillation (CRD) to enforce consistency in BEV representations by comparing the predictions of the teacher and the aligned student. We evaluate our method on the nuScenes dataset. Comprehensive experiments show that our method significantly improves camera-based BEV models and outperforms recent cross-modal knowledge distillation techniques.

1 Introduction

Camera-based bird's-eye-view (BEV) representations have gained significant attention in autonomous driving, from conventional perception systems [15, 21, 23, 34] to recent planning-oriented end-to-end systems [13, 38]. Leveraging the cost-effectiveness and rich semantic information of camera sensors, numerous attempts focus on learning accurate BEV representations from multi-view camera images to unify spatial and semantic information in surrounding environments. However, despite remarkable progress, a considerable performance gap remains between camera-based and LiDAR-based BEV models [3, 35] in several downstream tasks such as detection, tracking and planning, primarily due to the inherent loss of geometric fidelity during the image formation process.

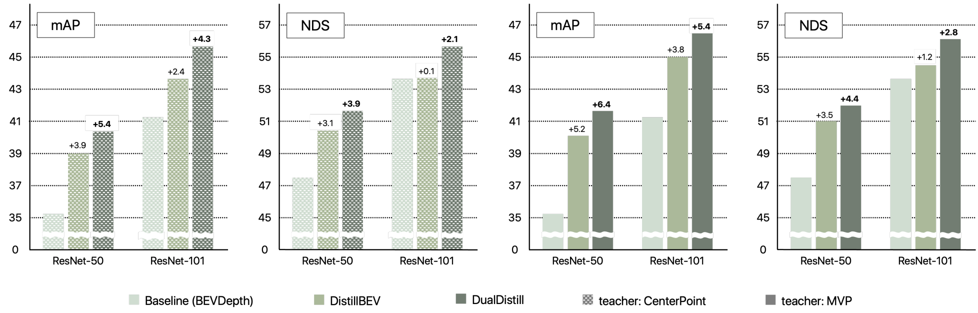


Figure 1: An overview of mAP and NDS improvements on the nuScenes [2] validation set. DualDistill achieves notable gains over its baseline [21] and outperforms the state-of-the-art DistillBEV [33], demonstrating the effectiveness of DualDistill.

Therefore, many studies [5, 6, 16, 18, 33, 39] have explored cross-modal knowledge distillation [12], where a LiDAR-based teacher model transfers precise geometric cues to a camera-based student. A common strategy is response-level distillation [27, 37], which treats the teacher's predictions as pseudo-labels. However, this approach can be suboptimal, as it relies solely on final outputs without fully exploiting the teacher's internal knowledge. An alternative is feature-level distillation [19, 33], where the student mimics the teacher's intermediate representations. However, feature-level distillation across modalities remains challenging due to the intrinsic differences between sensors: LiDAR provides accurate but sparse geometry, whereas cameras offer dense semantic context. Consequently, the LiDAR model may not serve as an ideal teacher [18], highlighting the need for proper alignment between student and teacher representations to balance the student's and the teacher's strengths.

In this work, we present an unified cross-modal knowledge distillation algorithm for 3D object detection, called DualDistill, to tackle the challenge of aligning camera and LiDAR BEV features. First, we propose an attention-guided orthogonal alignment (AOA) module that samples student features for distillation and transforms them into the teacher's space while preserving the student's inherent structure. To achieve this, AOA consists of a cross-attention projector and an orthogonal projector. Second, we integrate our AOA into a multi-scale feature distillation framework. In addition, we adopt an adaptive region weighting to distill knowledge from the imperfect teacher based on its importance. Third, we introduce cross-head response distillation (CRD), which enforces consistency by comparing the teacher's detection head output using the aligned student's BEV feature and the teacher's own BEV feature. We evaluate our method on the standard nuScenes [2] benchmark. Extensive experiments demonstrate that our approach significantly enhances camera-based BEV models, surpassing recent cross-modal knowledge distillation methods.

Our main contributions are outlined as follows:

- We propose attention-guided orthogonal alignment, which preserves the student's inherent feature structure while transforming it into the teacher's space, thereby mitigating modality-specific differences.

- We present cross-head response distillation to enhance cross-modal consistency by comparing the predictions of the shared detection head using both the teacher's and the aligned student's BEV representations.

- We introduce DualDistill, a framework that integrates AOA into multi-scale feature distillation and CRD into response-level distillation, enabling effective cross-modal knowledge transfer between camera and LiDAR modalities.
- The proposed DualDistill significantly improves the camera-based BEV models and outperforms recent state-of-the-art methods in cross-modal knowledge distillation on the nuScenes dataset.

2 Related Work

2.1 Camera-Based BEV Representation for 3D Object Detection

A camera-based bird's-eye-view (BEV) representation is a top-down feature map derived from multi-view images, encoding both spatial structure and semantic information of driving scenes. Due to its representational advantages, several studies [9, 14, 15, 23, 29, 30, 34] have explored camera-based BEV representations. LSS [30] lifts image features into 3D space. BEVDet [15] optimizes feature alignment and data augmentation. BEVFormer [23] and its extension [34] incorporate spatial and temporal consistency. BEVDet4D [14] aggregates multi-frame information to capture temporal dependencies. SOLOFusion [29] constructs historical BEV features to enhance temporal modeling. VideoBEV [9] alleviates the computational cost of SOLOFusion with recurrent modeling.

Some approaches [20, 21, 22] have explored incorporating additional depth supervision to enhance camera-based BEV representations. For instance, BEVDepth [21] introduces explicit depth supervision to improve BEV features. BEVStereo [20] utilizes temporal cues to filter out irrelevant depth candidates, selects the most probable ones to reduce computational cost, and refines them using the Expectation-Maximization (EM) algorithm [28]. BEVNeXt [22] enhances depth quality by integrating color cues and expands the receptive field to capture broader spatial context in dynamic 3D scenes. Despite these advances, camera-based methods still lag behind LiDAR-based approaches [3, 4, 35]. As a result, knowledge distillation [12] has emerged as a promising direction for further improving camera-based BEV perception.

2.2 Cross-Modal Knowledge Distillation for 3D Object Detection

Cross-modal knowledge distillation [8, 17, 32] transfers knowledge across sensor modalities, allowing a model trained on one modality to guide another. For example, early work [8] transfers supervision from labeled RGB images to unlabeled depth and optical flow, showing that mid-level features from a source modality can guide learning in a target modality. Subsequent studies extended this idea to heterogeneous settings. VGSR [17] distills visual knowledge into speak recognition models, and XKD [32] bridges audio and video modalities via domain alignment.

In autonomous driving, numerous cross-modal knowledge distillation methods [5, 6, 16, 18, 19, 33] have been proposed to improve camera-based student models by leverage depth-aware supervision from LiDAR-based teacher models. MonoDistill [6] aligns LiDAR signals with RGB images and transfer the spatial cues to the monocular 3D detector. BEVDistill [5] extends this to multi-camera 3D object detection by projecting features into the BEV space, which allows natural alignment between camera-based and LiDAR-based representations. TiG-BEV [16] proposes the target inner-geometry learning to alleviate the BEV feature gap

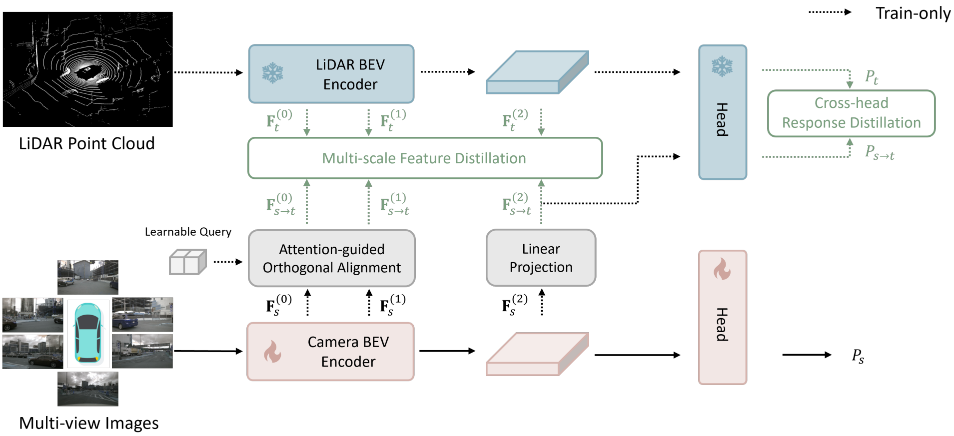


Figure 2: An overall architecture of DualDistill.

between two modalities. X^3KD [19] introduces a cross-modal and cross-task knowledge distillation framework for LiDAR to camera knowledge distillation. DistillBEV [33] introduces a region decomposition strategy to balance feature transfer across spatial regions. LabelDistill [18] mitigates the adverse effects of LiDAR supervision due to different sensor natures by integrating aleatoric uncertainty-free features from ground-truth labels.

3 Proposed Method

Figure 2 provides an overview of DualDistill, where the student is a camera-based model and the teacher is a LiDAR-based model. First, we propose the attention-guided orthogonal alignment (AOA) to address the discrepancy between sensor modalities. Second, we present the multi-scale feature distillation (MFD) based on the adaptive region weights [33] to effectively distill the teacher's representations. Third, we introduce the cross-head response distillation (CRD) to enforce consistency in the teacher's predictions across both the aligned student's and teacher's BEV representations. Finally, we train the student by simultaneously minimizing the detection loss, the multi-scale feature distillation loss, and the cross-head response distillation loss.

3.1 Attention-Guided Orthogonal Alignment

LiDAR-based models are not ideal teachers due to the inherent limitations of LiDAR sensors. Although LiDAR point clouds provide accurate depth information, their low resolution and lack of color reduce effectiveness in certain scenarios. Thus, successful knowledge distillation between camera and LiDAR modalities requires selectively transferring useful information while preserving critical student features. The AOA module addresses this by aligning student BEV features with teacher features via cross-attention and orthogonal projectors.

Cross-attention projector. Let $\mathbf{F}_s \in \mathbb{R}^{c_s \times h_s \times w_s}$ and $\mathbf{F}_t \in \mathbb{R}^{c_t \times h_t \times w_t}$ denote the BEV representations of the student and teacher, where c, h, and w indicate the number of channels, height, and width. We introduce learnable query embeddings $\mathbf{Q} \in \mathbb{R}^{c_s \times h_s \times w_s}$ that serve as feature samplers to extract relevant student features for comparison with the teacher's representations. Cross-attention is applied between the query embeddings and the student's BEV features, followed by layer normalization, a feed-forward network, and bilinear interpolation to ensure spatial alignment with the teacher's BEV features. Therefore, the output of the cross-attention projector $\tilde{\mathbf{F}}_{s \to t} \in \mathbb{R}^{c_s \times h_t \times w_t}$ is given by

$$\tilde{\mathbf{F}}_{s \to t} = \texttt{Upsample}(\texttt{FFN}(\texttt{LN}(\texttt{Attn}(\mathbf{Q}, \mathbf{F}_s, \mathbf{F}_s)))) \tag{1}$$

where $\texttt{Attn}(\cdot,\cdot,\cdot)$, $\texttt{LN}(\cdot)$, $\texttt{FFN}(\cdot)$, and $\texttt{Upsample}(\cdot)$ denote window-based attention [24], layer normalization [1], feed-forward network, and bilinear interpolation, respectively.

Orthogonal projector. The orthogonal projector [26] aligns the student's BEV features with those of the teacher while preserving their structural consistency. To enforce the orthogonality constraint, we re-parameterize the learnable projection matrix $\mathbf{P}$ using the matrix exponential, implemented via Padé approximation [11]:

$$\tilde{\mathbf{P}} = \arg\min_{\mathbf{P}} \|\mathbf{P}^T \mathbf{P} - \mathbf{I}\|_F \tag{2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and $\tilde{\mathbf{P}}$ is the resulting orthogonal projection matrix. The aligned student BEV features $\mathbf{F}_{s \to t} \in \mathbb{R}^{c_t \times h_t \times w_t}$ are then obtained by multiplying $\tilde{\mathbf{P}}$ with the cross-attention projector's output, *i.e.*, $\mathbf{F}_{s \to t} = \tilde{\mathbf{P}} \tilde{\mathbf{F}}_{s \to t}$. Orthogonal projection is particularly effective for cross-modal distillation, as it preserves the inner product of the student features. This helps maintain their structure and mitigates imperfections in the teacher model.

3.2 Multi-Scale Feature Distillation

Adaptive region weighting is widely used to distill BEV representations between camera- and LiDAR-based models. It identifies informative regions where the student should mimic the teacher and scales the penalty on feature discrepancies according to regional significance. Following prior work [33], we estimate region importance from region categories and salient pixels, and use them to define the multi-scale feature distillation loss.

Category-aware region weights. The BEV space is divided into four categories: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). TP and FN contain objects, while FP refers to regions where the teacher assigns high confidence, and TN is considered relatively less important. We assign distinct weights to each category. Specifically, the category-aware region weight C at pixel (i, j) is defined as

$$C(i,j) = \begin{cases} \frac{\alpha}{N_{\mathrm{TP}} + N_{\mathrm{FN}}}, & \text{if } (i,j) \in \mathrm{TP} \text{ or } \mathrm{FN} \\ \frac{\beta}{N_{\mathrm{FP}}}, & \text{if } (i,j) \in \mathrm{FP} \\ \frac{\gamma}{N_{\mathrm{TN}}}, & \text{if } (i,j) \in \mathrm{TN} \end{cases} \tag{3}$$

where $N_{\mathrm{TP}}, N_{\mathrm{FN}}, N_{\mathrm{FP}}, N_{\mathrm{TN}}$ denote the number of pixels in each category, and α, β and γ control their relative importance.

Saliency-aware region weights. We also consider the response magnitudes of both student and teacher models. Given their BEV features, we compute the mean absolute values across channels and normalize them via softmax. The teacher's saliency-aware region weights are defined as $S_t = N_{\mathbf{F}_t} \cdot \texttt{softmax}(\bar{\mathbf{F}}_t/\omega)$, where $\bar{\mathbf{F}}_t$ is the mean absolute feature map, ω is a temperature parameter, and $N_{\mathbf{F}_t}$ denotes the number of BEV pixels, ensuring consistent scaling. The student's weights S_s are computed in the same way, and the final saliency-aware region weights are obtained by averaging: $S = \frac{1}{2}(S_t + S_s)$.

Multi-scale feature distillation. A multi-scale feature distillation loss encourages the student to imitate the teacher's representations across multiple layers while focusing on informative regions. Let $\mathbf{F}^{(n)}_{s\to t}$ and $\mathbf{F}^{(n)}_t$ denote the aligned student and teacher BEV features at scale $n \in \{0,1,2\}$. As shown in Figure 2, $n = 0,1$ correspond to intermediate BEV features, and $n = 2$ represents the final BEV feature before the task head. We align the intermediate features $\mathbf{F}^{(0)}_s$, $\mathbf{F}^{(1)}_s$ using the AOA module, while the final feature $\mathbf{F}^{(2)}_s$ is mapped via a linear projection. As done in [33], we use a linear projection at $n = 2$ since both teacher and student already capture similar task-level semantics. We compute the feature difference as $\mathbf{D}^{(n)} = \mathbf{F}^{(n)}_t - \mathbf{F}^{(n)}_{s\to t}$, and apply category-aware weights $C^{(n)}$ and saliency-aware weights $S^{(n)}$ to obtain the loss:

$$\mathcal{L}_{\text{mfd}} = \sum_{n=0}^{2} \left\| \mathbf{C}^{(n)} \odot \mathbf{S}^{(n)} \odot \mathbf{D}^{(n)} \right\|_F \tag{4}$$

where $\odot$ denotes element-wise multiplication, and $\mathbf{C}^{(n)}$, $\mathbf{S}^{(n)}$ are reshaped for broadcasting. These weights highlight discrepancies in important regions while suppressing less critical ones.

3.3 Cross-Head Response Distillation

Cross-head response distillation transfers the teacher's depth-aware knowledge to the student by comparing their predictions using a shared detection head. As illustrated in Figure 2, the aligned student BEV feature $\mathbf{F}^{(2)}_{s\to t}$ is fed into the pre-trained teacher head $\texttt{Head}_t(\cdot)$ to produce cross-head predictions, *i.e.*, $P_{s\to t} = \texttt{Head}_t(\mathbf{F}^{(2)}_{s\to t})$. The teacher predictions are obtained from its own BEV features. However, low-confidence predictions from the teacher can introduce noise and hinder learning. To mitigate this, we refine the teacher logits via entropy minimization [7], resulting in $P_t = \texttt{Head}_t(\mathbf{F}^{(2)}_t)/\tau$, where τ is a temperature parameter to amplify confident predictions and suppress uncertain ones. Finally, we define the cross-head distillation loss as

$$\mathcal{L}_{\text{crd}} = \mathcal{L}_{\text{det}}(P_t, P_{s\to t}) \tag{5}$$

where $\mathcal{L}_{\text{det}}$ is detection loss of BEVDepth [21], consisting of classification and bounding box regression terms.

3.4 Training Objective

Our training objective comprises three loss functions:

$$\mathcal{L} = \mathcal{L}_{\text{det}}(P_{gt}, P_s) + \mathcal{L}_{\text{mfd}} + \mathcal{L}_{\text{crd}} \tag{6}$$

where P_{gt} denotes the ground-truth detection boxes and $\mathcal{L}_{\text{det}}$ is the detection loss [21] to penalize the discrepancy between predictions and ground-truth. Also, $\mathcal{L}_{\text{mfd}}$ is the the multi-scale feature distillation loss in (4), and $\mathcal{L}_{\text{crd}}$ is the cross-head response distillation loss in (5).

4 Experiments

In this section, we describe the experiment setup including dataset, evaluation metrics, and implementation details. We compare our method with state-of-the-arts. We then perform ablation studies to analyze our DualDistill.

4.1 Experimental Setup

Dataset and metrics. We perform experiments on the nuScenes dataset [2], which contains 1,000 driving sequences with 3D object annotations at 2Hz. It includes 1.4M annotated object instances across 40K keyframes, captured by six cameras and a 32-beam LiDAR. We primarily use mean average precision (mAP) and nuScenes detection score (NDS) as evaluation metrics.

Implementation details. We use BEVDepth [21] as the student model with ResNet-50 and ResNet-101 backbones [10] pre-trained on ImageNet [31]. We adopt CenterPoint [35] and MVP [36] as teacher models. We train the student for 24 epochs with a batch size of 32 using the AdamW optimizer [25] with a weight decay of 0.01. The learning rate is initialized to 2×10^{-4} and decayed by a factor of 0.1 at epochs 16 and 22. Unless otherwise specified, the input resolution is set to 256×704, and we follow the data augmentation strategies in [15, 21]. We emprically set our hyperparameters to $\alpha = 6 \times 10^{-3}, \beta = 4 \times 10^{-2}, \gamma = 6 \times 10^{-2}, \omega = 0.5$ and $\tau = 0.1$., respectively.

4.2 Main Results

Table 1 compares our method with recent state-of-the-art techniques [16, 18, 19, 33] in cross-modal knowledge distillation. Using the CenterPoint teacher, our method significantly outperforms the baseline, BEVDepth [21], improving mAP and NDS by 5.4 and 3.9 points with ResNet-50, and by 4.3 and 2.1 points with ResNet-101. With the MVP teacher, the gains are even larger: 6.4 and 4.4 points with ResNet-50, and 5.4 and 2.8 points with ResNet-101. Note that these performance gains require 15% additional training-time parameters in the distillation modules (BEVDepth: 53.2M vs. ours: 61.5M), which are discarded after training, keeping inference-time parameters and computation identical to the baseline. In addition, our method achieves competitive or superior results compared to existing methods, ranking second with ResNet-50 and first with ResNet-101. Notably, while LabelDistill shows limited scalability with larger backbones, our method maintains strong and consistent performance.

A comparison with DistillBEV [33], which adopts adaptive region weighting for multi-scale feature distillation, highlights the effectiveness of AOA and CRD. In Table 1, our method consistently outperforms DistillBEV across different backbones and teacher models. When we use CenterPoint teacher, mAP and NDS improve by 1.5 and 0.8 points with

Model	Base	Teacher	Img Size	mAP ↑ (Δ)	NDS ↑ (Δ)
CenterPoint [35]	–	–	–	56.4	64.8
MVP [36]	–	–	–	67.1	78.0
Backbone: ResNet-50					
BEVDepth [21]	–	–	256×704	35.1	47.5
TiG-BEV† [16]	BEVDepth	CenterPoint	256×704	36.6 (+1.5)	46.1 (–1.4)
X³KD [19]	BEVDepth	CenterPoint	256×704	39.0 (+3.9)	50.5 (+3.0)
DistillBEV [33]	BEVDepth	CenterPoint	256×704	39.0 (+3.9)	50.6 (+3.1)
LabelDistill† [18]	BEVDepth	CenterPoint	256×704	**41.9 (+6.8)**	**52.8 (+5.3)**
Ours	BEVDepth	CenterPoint	256×704	40.5 (+5.4)	51.4 (+3.9)
DistillBEV [33]	BEVDepth	MVP	256×704	40.3 (+5.2)	51.0 (+3.5)
Ours	BEVDepth	MVP	256×704	**41.5 (+6.4)**	**51.9 (+4.4)**
Backbone: ResNet-101					
BEVDepth [21]	–	–	512×1408	41.2	53.5
TiG-BEV [16]	BEVDepth	CenterPoint	512×1408	44.0 (+2.8)	54.4 (+0.9)
X³KD [19]	BEVDepth	CenterPoint	512×1408	44.8 (+3.6)	55.3 (+1.8)
DistillBEV [33]	BEVDepth	CenterPoint	512×1408	43.6 (+2.4)	53.6 (+0.1)
LabelDistill† [18]	BEVDepth	CenterPoint	512×1408	45.1 (+3.9)	55.3 (+1.8)
Ours	BEVDepth	CenterPoint	512×1408	**45.5 (+4.3)**	**55.6 (+2.1)**
DistillBEV [33]	BEVDepth	MVP	512×1408	45.0 (+3.8)	54.7 (+1.2)
Ours	BEVDepth	MVP	512×1408	**46.6 (+5.4)**	**56.3 (+2.8)**

Table 1: Comparison to other LiDAR-guided cross-modal knowledge distillation strategies on the nuScenes [2] validation set. The best results are depicted in bold and the second-best results are underlined. Our method is highlighted in gray. † denote methods without CBGS [40].

ResNet-50, and by 1.9 and 2.0 points with ResNet-101. In MVP teacher setting, the gains are 1.2 and 0.9 points with ResNet-50, and 1.6 and 1.6 points with ResNet-101.

Figure 3 illustrates qualitative results of DualDistill and BEVDepth baseline. In Figure 3, our method successfully identifies distant objects that the baseline fails to capture and avoids misclassifications such as containers for vehicles, as indicated by the red arrows. Additionally, it achieves more precise localization, refining object positioning and depth estimation compared to the baseline, as indicated by the green arrows.

4.3 Ablation Study

Distillation strategy. Table 2 (a) reports the performance under different combinations of distillation strategies. Compared to BEVDepth baseline (*i.e.*, without MFD and CRD), applying MFD improves mAP and NDS by 5.3 and 3.8 points, respectively. This shows that MFD enables the student to better mimic the teacher's BEV representations while preserving structural consistency and enhancing feature transfer through adaptive region weighting. Adding CRD yields an additional gain of 1.1 mAP and 0.6 NDS by transferring depth-aware knowledge and improving the student's depth reasoning. These results confirm the effectiveness of both distillation strategies, as their combination yields significant improvements over the baseline.

Alignment strategy. Table 2 (b) evaluates alignment strategies at intermediate distillation layers. We compare our proposed AOA with the projection block in DistillBEV [33], which

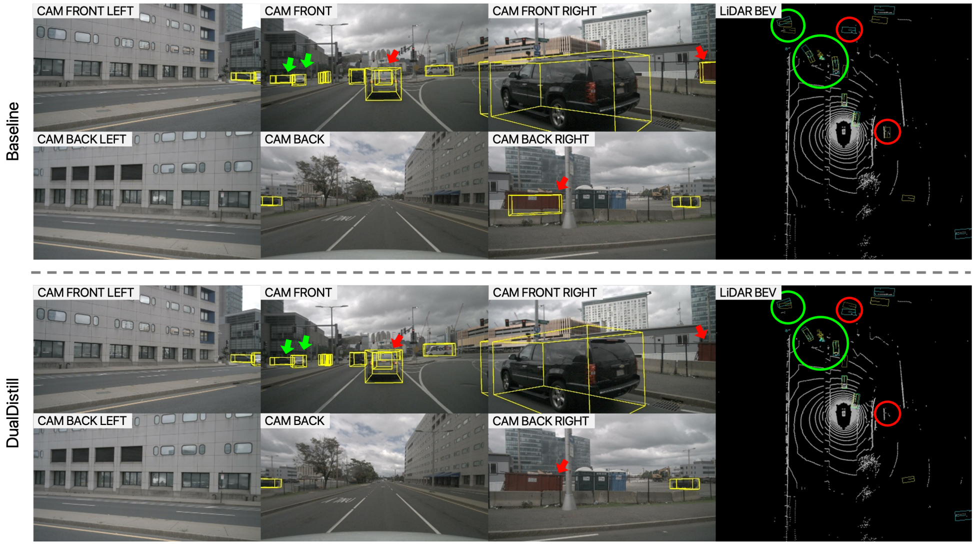


Figure 3: Qualitative results comparing DualDistill with the baseline [21]. Sky-blue boxes mark ground-truth objects; yellow boxes show detections. Red arrows highlight baseline misdetections corrected by DualDistill, and green arrows indicate improved localization.

(a)

MFD	CRD	mAP	NDS
		35.1	47.5
✓		40.4	51.3
✓	✓	41.5	51.9

(b)

Alignment	mAP	NDS
Upsample - Conv×3	41.0	51.3
AOA	41.5	51.9

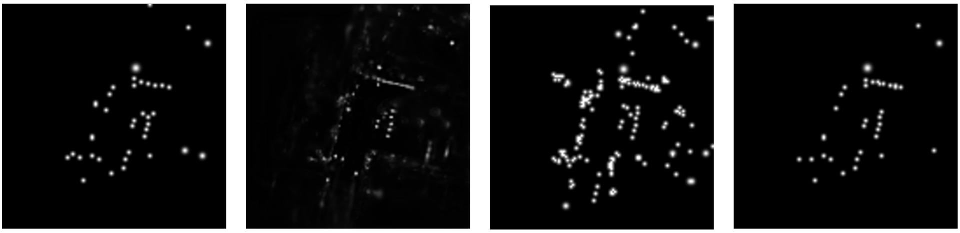
(c)

Labels	τ	mAP	NDS
GT	–	40.3	51.1
Teacher	1.0	40.9	51.5
	0.5	41.2	51.9
	0.1	41.5	51.9

Table 2: Ablation study on DualDistill: (a) compares different distillation strategies, (b) evaluates alignment strategies, and (c) examines the impact of pseudo-labeling and temperature settings. Our method is highlighted in gray.

combines bilinear upsampling with three stacked convolution layers. At the final distillation layer, both methods adopt a 1×1 convolution, following DistillBEV's finding that simple linear alignment is sufficient at this stage, while intermediate layers require more complex alignment due to the larger modality gap. As shown in Table 2 (b), AOA improves mAP and NDS by 0.5 and 0.6 points over the projection block in DistillBEV, indicating its effectiveness in aligning camera and LiDAR features and mitigating the modality gap.

Pseudo-labels. We analyze the impact of pseudo-labeling in CRD. Specifically, we compare ground-truth (GT) and teacher predictions as supervision signals to examine whether teacher predictions offer better guidance than GTs. As shown in Table 2 (c), using teacher predictions yields higher performance, improving mAP and NDS by 0.6 and 0.4 points, respectively. This suggests that teacher predictions are more effective supervision signals, as



(a) GT (b) Predictions (c) τ=1 (d) τ=0.1

Figure 4: Comparison of label heatmaps. The brightness of each heatmap represents confidence. Ground-truth (GT) is generated by applying a Gaussian kernel at annotated centers. Pseudo-label (τ=1 and τ=0.1) are obtained by thresholding the teacher's predictions to identify valid objects, followed by applying a Gaussian kernel at the object centers.

they convey how to better exploit BEV representations for improved inference.

Temperature scaling. We examine the effect of temperature scaling on teacher predictions used for pseudo-label generation. By tuning the temperature parameter τ, we control the sharpness of the confidence distribution, amplifying reliable predictions while suppressing uncertainty. As shown in Figure 4, decreasing τ progressively filters out ambiguous predictions, yielding pseudo-labels that concentrate on high-confidence detections. Table 2 reports performance across different τ settings and shows that lower τ consistently improves results. These indicate that sharpening the confidence distribution enables the student to focus on reliable geometric cues, boosting object detection performance.

5 Conclusion

In this work, we presented DualDistill, a unified cross-modal knowledge distillation framework designed to enhance camera-based bird's-eye-view (BEV) models by effectively distilling knowledge from a LiDAR-based teacher. First, we proposed attention-guided orthogonal alignment (AOA) module, which aligns student features with the teacher's representations while preserving student's critical structural information. We then incorporated AOA module into a multi-scale feature distillation framework with adaptive region weighting scheme. We also presented cross-head response distillation (CRD) that compares the predictions from teacher's and aligned student's features, to enforce consistency in BEV representations. We conducted extensive experiments on the nuScenes dataset. Experimental results validated the superiority of our approach, showing that it surpasses recent state-of-the-art cross-modal knowledge distillation techniques.

Acknowledgements

This work was supported in part by the National Research Foundation of Korea (NRF) funded by the Korean Goverment under Grants RS-2023-00221365 and RS-2024-00352566.

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

[2] Holger Caesar, Varun Bankiti, Alex H Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multimodal dataset for autonomous driving. In *CVPR*, pages 11621–11631, 2020.

[3] Yilun Chen, Zhiding Yu, Yukang Chen, Shiyi Lan, Anima Anandkumar, Jiaya Jia, and Jose M Alvarez. Focalformer3d: focusing on hard instance for 3d object detection. In *ICCV*, pages 8394–8405, 2023.

[4] Yukang Chen, Jianhui Liu, Xiangyu Zhang, Xiaojuan Qi, and Jiaya Jia. Voxelnext: Fully sparse voxelnet for 3d object detection and tracking. In *CVPR*, pages 21674–21683, 2023.

[5] Zehui Chen, Zhenyu Li, Shiquan Zhang, Liangji Fang, Qinhong Jiang, and Feng Zhao. Bevdistill: Cross-modal bev distillation for multi-view 3d object detection. In *ICLR*, 2023.

[6] Zhiyu Chong, Xinzhu Ma, Hong Zhang, Yuxin Yue, Haojie Li, Zhihui Wang, and Wanli Ouyang. Monodistill: Learning spatial features for monocular 3d object detection. In *ICLR*, 2022.

[7] Yves Grandvalet and Yoshua Bengio. Semi-supervised learning by entropy minimization. In *NeurIPS*, volume 17, 2004.

[8] Saurabh Gupta, Judy Hoffman, and Jitendra Malik. Cross modal distillation for supervision transfer. In *CVPR*, June 2016.

[9] Chunrui Han, Jinrong Yang, Jianjian Sun, Zheng Ge, Runpei Dong, Hongyu Zhou, Weixin Mao, Yuang Peng, and Xiangyu Zhang. Exploring recurrent long-term temporal fusion for multi-view 3d perception. *IEEE Robotics and Automation Letters*, 9(7): 6544–6551, 2024.

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.

[11] Nicholas J Higham. The scaling and squaring method for the matrix exponential revisited. *SIMAX*, 26(4):1179–1193, 2005.

[12] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[13] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *CVPR*, pages 17853–17862, 2023.

[14] Junjie Huang and Guan Huang. Bevdet4d: Exploit temporal cues in multi-camera 3d object detection. *arXiv preprint arXiv:2203.17054*, 2022.

[15] Junjie Huang, Guan Huang, Zheng Zhu, Yun Ye, and Dalong Du. Bevdet: High-performance multi-camera 3d object detection in bird-eye-view. *arXiv preprint arXiv:2112.11790*, 2021.

[16] Peixiang Huang, Li Liu, Renrui Zhang, Song Zhang, Xinli Xu, Baichao Wang, and Guoyi Liu. Tig-bev: Multi-view bev 3d object detection via target inner-geometry learning. *arXiv preprint arXiv:2212.13979*, 2022.

[17] Yufeng Jin, Guosheng Hu, Haonan Chen, Duoqian Miao, Liang Hu, and Cairong Zhao. Cross-modal distillation for speaker recognition. In *AAAI*, volume 37, pages 12977–12985, June 2023.

[18] Sanmin Kim, Youngseok Kim, Sihwan Hwang, Hyeonjun Jeong, and Dongsuk Kum. Labeldistill: Label-guided cross-modal knowledge distillation for camera-based 3d object detection. In *ECCV*, pages 19–37. Springer, 2024.

[19] Marvin Klingner, Shubhankar Borse, Varun Ravi Kumar, Behnaz Rezaei, Venkatraman Narayanan, Senthil Yogamani, and Fatih Porikli. X3kd: Knowledge distillation across modalities, tasks and stages for multi-camera 3d object detection. In *CVPR*, pages 13343–13353, 2023.

[20] Yinhao Li, Han Bao, Zheng Ge, Jinrong Yang, Jianjian Sun, and Zeming Li. Bevstereo: Enhancing depth estimation in multi-view 3d object detection with temporal stereo. In *AAAI*, volume 37, pages 1486–1494, 2023.

[21] Yinhao Li, Zheng Ge, Guanyi Yu, Jinrong Yang, Zengran Wang, Yukang Shi, Jianjian Sun, and Zeming Li. Bevdepth: Acquisition of reliable depth for multi-view 3d object detection. In *AAAI*, volume 37, pages 1477–1485, 2023.

[22] Zhenxin Li, Shiyi Lan, Jose M Alvarez, and Zuxuan Wu. Bevnext: Reviving dense bev frameworks for 3d object detection. In *CVPR*, pages 20113–20123, 2024.

[23] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. Bevformer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *ECCV*, pages 1–18, 2022.

[24] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, pages 10012–10022, 2021.

[25] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.

[26] Roy Miles, Ismail Elezi, and Jiankang Deng. Vkd: Improving knowledge distillation using orthogonal projections. In *CVPR*, pages 15720–15730, 2024.

[27] Seyed Iman Mirzadeh, Mehrdad Farajtabar, Ang Li, Nir Levine, Akihiro Matsukawa, and Hassan Ghasemzadeh. Improved knowledge distillation via teacher assistant. In *AAAI*, volume 34, pages 5191–5198, 2020.

[28] Todd K Moon. The expectation-maximization algorithm. *IEEE SPM*, 13(6):47–60, 1996.

[29] Jinhyung Park, Chenfeng Xu, Shijia Yang, Kurt Keutzer, Kris M. Kitani, Masayoshi Tomizuka, and Wei Zhan. Time will tell: New outlooks and a baseline for temporal multi-view 3d object detection. In *ICLR*, 2023.

[30] Jonah Philion and Sanja Fidler. Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d. In *ECCV*, 2020.

[31] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *IJCV*, 115:211–252, 2015.

[32] Pritam Sarkar and Ali Etemad. Xkd: Cross-modal knowledge distillation with domain alignment for video representation learning. In *AAAI*, volume 38, pages 14875–14885, March 2024.

[33] Zeyu Wang, Dingwen Li, Chenxu Luo, Cihang Xie, and Xiaodong Yang. Distillbev: Boosting multi-camera 3d object detection with cross-modal knowledge distillation. In *ICCV*, pages 8637–8646, 2023.

[34] Chenyu Yang, Yuntao Chen, Hao Tian, Chenxin Tao, Xizhou Zhu, Zhaoxiang Zhang, Gao Huang, Hongyang Li, Yu Qiao, Lewei Lu, et al. Bevformer v2: Adapting modern image backbones to bird's-eye-view recognition via perspective supervision. In *CVPR*, pages 17830–17839, 2023.

[35] Tianwei Yin, Xingyi Zhou, and Philipp Krahenbuhl. Center-based 3d object detection and tracking. In *CVPR*, pages 11784–11793, 2021.

[36] Tianwei Yin, Xingyi Zhou, and Philipp Krähenbühl. Multimodal virtual point 3d detection. In *NeurIPS*, volume 34, pages 16494–16507, 2021.

[37] Borui Zhao, Quan Cui, Renjie Song, Yiyu Qiu, and Jiajun Liang. Decoupled knowledge distillation. In *CVPR*, pages 11953–11962, 2022.

[38] Wenzhao Zheng, Ruiqi Song, Xianda Guo, Chenming Zhang, and Long Chen. Genad: Generative end-to-end autonomous driving. In *ECCV*, pages 87–104. Springer, 2024.

[39] Shengchao Zhou, Weizhou Liu, Chen Hu, Shuchang Zhou, and Chao Ma. Unidistill: A universal cross-modality knowledge distillation framework for 3d object detection in bird's-eye view. In *CVPR*, pages 5116–5125, 2023.

[40] Benjin Zhu, Zhengkai Jiang, Xiangxin Zhou, Zeming Li, and Gang Yu. Class-balanced grouping and sampling for point cloud 3d object detection. *arXiv preprint arXiv:1908.09492*, 2019.